The stated system-of-trivial-centralizers question is universally affirmative

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Abstract

The first concluding question of Cabello Sánchez's paper asks whether each $\mathcal{M}$-centralizer $\Omega : L_0^p \to L^q$, $0 < q < p < \infty$, admits a system of trivial centralizers whose distances from the module morphisms have supremum equal to the corresponding distance of Ω. As written, the question places no relation between Ω and the proposed system. We prove that this omission makes the answer universally affirmative for arbitrary quasinormed modules and all exponents. If Ω has finite distance, use Ω itself. If its distance is infinite, scale one bounded nonlinear homogeneous map having positive distance from the morphisms. Such maps are trivial centralizers. This resolves the literal display but does not produce the intended analogue of the paper's finite-rank localization lemma.

1 Exact source question

The source asks the following in its concluding remarks:

> Most results in Sections 3 and 4 would generalize to noncommutative L^p spaces associated to arbitrary von Neumann algebras as long as one could find a good substitute for Lemma 3. More precisely, we ask if for every $\mathcal{M}$-centralizer $\Omega : L_0^p \to L^q$ with $0 < q < p < \infty$ there is a system of trivial centralizers Ω_i such that
>
> $$\operatorname{dist}(\Omega, \mathscr{M}_{\mathcal{M}}(L_0^p, L^q)) = \sup_i \operatorname{dist}(\Omega_i, \mathscr{M}_{\mathcal{M}}(L_0^p, L^q)).$$
>
> Here, $L_0^p = \{af^{1/p} : a \in \mathcal{M}\}$, where f is a normal, faithful state on $\mathcal{M}$.

The exact source page is reproduced below.

8. Concluding remarks

★ Most results in Sections 3 and 4 would generalize to noncommutative L^p spaces associated to arbitrary von Neumann algebras as long as one could find a good substitute for Lemma 3. More precisely, we ask if for every $\mathcal{M}$-centralizer $\Omega : L_0^p \longrightarrow L^q$ with $0 < q < p < \infty$ there is a system of trivial centralizers Ω_i such that $\operatorname{dist}(\Omega, \mathcal{M}_{\mathcal{M}}(L_0^p, L^q)) = \sup_i \operatorname{dist}(\Omega_i, \mathcal{M}_{\mathcal{M}}(L_0^p, L^q))$. Here, $L_0^p = \{af^{1/p} : a \in \mathcal{M}\}$, where f is a normal, faithful state on $\mathcal{M}$.

The preceding Lemma 3 uses the concrete localizations $\Phi_e(x) = \Phi(xe)$ for finite-rank projections e. The displayed question, however, does not say that Ω_i must be a localization of Ω, nor does it impose any compatibility, convergence, or uniform centralizer bound. The proof below takes the quantifiers exactly as displayed.

2 Definitions and proof intuition

For homogeneous maps $F : X \to Y$ between quasinormed spaces, set

$$\|F\|_{\mathrm{h}} = \sup_{x\neq 0} \frac{\|F(x)\|_Y}{\|x\|_X} \quad \text{and} \quad \operatorname{dist}(F, \mathscr{N}) = \inf_{T\in\mathscr{N}} \|F - T\|_{\mathrm{h}},$$

with values in $[0, \infty]$. Here $\mathscr{N}$ is the vector space of algebraic module morphisms. A centralizer is trivial precisely when its distance from $\mathscr{N}$ is finite.

The key observation is numerical. The question asks for a family of trivial maps realizing the same extended number $D = \operatorname{dist}(\Omega, \mathscr{N})$, but it does not require those maps to contain any information about Ω. Finite D is realized by Ω itself. Infinite D is realized by scalar multiples of one unrelated trivial map whose distance is positive.

3 A trivial centralizer at positive distance

We use only the standard assumptions that the module actions are jointly continuous and that the quasi-norms are homogeneous and Hausdorff.

Lemma 1. *Let X and Y be nonzero complex quasinormed left modules over a Banach algebra A, with $\dim_{\mathbb{C}} X > 1$. Let $\mathscr{N} = \mathscr{M}_A(X, Y)$. There is a bounded homogeneous map $B : X \to Y$ such that:*

1. *B is quasilinear and an A-centralizer;*
2. *B is a trivial centralizer;*
3. $0 < \operatorname{dist}(B, \mathscr{N}) < \infty$.

Proof. Choose a unit representative s_L on each complex one-dimensional subspace $L \subset X$. Fix two linearly independent unit vectors u, v, and choose $s_{\mathbb{C}u} = u$. Every nonzero $x \in X$ has a unique representation $x = \lambda s_L$. Define

$$b(0) = 0, \qquad b(\lambda s_L) = \begin{cases} \lambda, & L = \mathbb{C}u, \\ 0, & L \neq \mathbb{C}u. \end{cases}$$

Then b is complex homogeneous and $|b(x)| \leq \|x\|_X$. It is not additive: $b(u) = 1$, $b(v) = 0$, whereas $b(u + v) = 0$. Fix a unit vector $y_0 \in Y$ and put $B(x) = b(x)y_0$. Thus B is bounded, homogeneous, and not additive.

Let K_X, K_Y be quasi-triangle constants and let C_X, C_Y bound the two module actions. Bounded homogeneity gives

$$\begin{aligned} \|B(x + z) - B(x) - B(z)\|_Y &\leq C\big(\|x + z\|_X + \|x\|_X + \|z\|_X\big) \\ &\leq C'\big(\|x\|_X + \|z\|_X\big), \end{aligned}$$

so B is quasilinear. Likewise,

$$\begin{aligned} \|B(ax) - aB(x)\|_Y &\leq C\big(\|B(ax)\|_Y + \|aB(x)\|_Y\big) \\ &\leq C'\|a\|_A\|x\|_X, \end{aligned}$$

so B is a left centralizer. Since B is bounded, the zero morphism witnesses triviality and $\operatorname{dist}(B,\mathscr{N}) \leq \|B\|_{\mathrm{h}} < \infty$.

It remains to exclude distance zero. If $\operatorname{dist}(B,\mathscr{N}) = 0$, for every n one can choose $T_n \in \mathscr{N}$ satisfying

$$\|(B - T_n)(x)\|_Y \leq n^{-1}\|x\|_X \qquad (x \in X).$$

Since T_n is additive, the quasi-triangle inequality yields, for fixed $x, z \in X$,

$$\|B(x+z) - B(x) - B(z)\|_Y \leq \frac{C}{n}\big(\|x+z\|_X + \|x\|_X + \|z\|_X\big).$$

Letting $n \to \infty$ forces $B(x+z) = B(x) + B(z)$, contradicting the construction. Hence $\operatorname{dist}(B,\mathscr{N}) > 0$. □

No local convexity is used; in particular, the lemma applies when the target is L^q with $0 < q < 1$.

4 Universal answer to the displayed question

Theorem 1. *Let X, Y be complex quasinormed modules and let $\mathscr{N} = \mathscr{M}_A(X,Y)$. For every centralizer $\Omega : X \to Y$, there exists a system of trivial centralizers (Ω_i) such that*

$$\operatorname{dist}(\Omega,\mathscr{N}) = \sup_i \operatorname{dist}(\Omega_i,\mathscr{N}).$$

Consequently, the first concluding question of arXiv:1910.08774 has an affirmative answer exactly as written, for every $p, q > 0$ and every von Neumann algebra $\mathcal{M}$.

Proof. Put $D = \operatorname{dist}(\Omega,\mathscr{N})$.

If $D < \infty$, then by definition there is a morphism at finite distance from Ω, so Ω is a trivial centralizer. Take the one-element system $\Omega_1 = \Omega$. Its supremal distance is exactly D.

Suppose $D = \infty$. Then Ω cannot be bounded, because the zero morphism would otherwise give finite distance. In particular X cannot be one-dimensional: every complex homogeneous map on a one-dimensional quasinormed space is bounded. Apply Lemma 1 and put

$$d = \operatorname{dist}(B,\mathscr{N}), \qquad \Theta = d^{-1}B.$$

The class $\mathscr{N}$ is a complex vector space, hence distance is homogeneous:

$$\operatorname{dist}(\lambda B,\mathscr{N}) = |\lambda| \operatorname{dist}(B,\mathscr{N}).$$

Therefore every $\Omega_n = n\Theta$ is a trivial centralizer and

$$\operatorname{dist}(\Omega_n,\mathscr{N}) = n.$$

It follows that $\sup_n \operatorname{dist}(\Omega_n,\mathscr{N}) = \infty = D$, as required. □

5 Why this is not the intended Lemma 3 substitute

The theorem is deliberately overgeneral: it uses neither $q < p$ nor any property of von Neumann algebras. This shows that the displayed formulation does not encode the local-to-global mechanism described immediately before it.

In the Schatten setting, the local maps are prescribed by $\Phi_e(x) = \Phi(xe)$, where e ranges over finite-rank projections. Right multiplication commutes with the left action, the compressed operator lies in the finite-rank core, and $\|x\|_\infty \le \|x\|_p$ makes every Φ_e trivial. An ultrafilter and WOT lower semicontinuity then recover the global distance.

For $L_0^p = \mathcal{M} f^{1/p}$, an arbitrary right projection need not preserve this core. Modular-analytic right multipliers can preserve it, but the elementary centralizer estimate controls $\|a\|_\infty$ rather than $\|af^{1/p}\|_p$, so the crucial Schatten bound still does not follow. The argument above says nothing about this obstruction.

A meaningful repaired question must specify, at minimum:

1. how every Ω_i is obtained from the given Ω;
2. which local pieces of L_0^p the index set exhausts;
3. the compatibility needed to patch the approximating morphisms; and
4. any required uniform bound on centralizer constants.

With such conditions imposed, Theorem 1 no longer applies, and the substantive arbitrary-von-Neumann-algebra problem remains unresolved here.

6 Verification and novelty audit

The proof reduces to four exact checks:

1. the chosen-line map b is complex homogeneous, bounded, and nonadditive;
2. bounded homogeneous maps are quasilinear centralizers and are trivial via zero;
3. zero distance from the morphisms would force additivity;
4. distance from a vector subspace is homogeneous under scalar dilation.

Each check is written out above and remains valid for quasi-norms.

Exact-phrase searches, arXiv searches for the displayed L_0^p question, the run's result indexes, and later papers citing the source did not locate a published correction or answer. This is evidence rather than a definitive citation audit. The mathematical claim is a full answer only to the literal display. Before dissemination, a specialist should verify that the author did not intend a convention in which the word "system" silently includes compression/localization data. If no such convention exists, the useful conclusion is that the question needs an additional hypothesis, not that the intended extension theory has been solved.

References

[1] F. Cabello Sánchez, *Nonlinear centralizers in homology II. The Schatten classes*, Rev. Mat. Iberoam. 37 (2021), no. 6, 2309–2346; arXiv:1910.08774.

[2] F. Cabello Sánchez, J. M. F. Castillo, S. Goldstein, and J. Suárez de la Fuente, *Twisting non-commutative L_p spaces*, Adv. Math. 294 (2016), 454–488; arXiv:1407.0969.